Translation, rotation, vibration

Q for ideal gases $\langle U \rangle = -\frac{\partial \ln Q}{\partial \beta}$ $\beta = \frac{1}{k_B T}$

$U_{vib} = \sum_i \left(\underbrace{\frac{1}{2} h\nu_i}_{\text{ZPE}} + \underbrace{\frac{h\nu_i}{e^{\beta h\nu_i} - 1}}_{\text{Thermal excitation}} \right)$ i = normal vib modes of molecules

$U_{rot} = \begin{cases} \frac{3}{2} k_B T & \text{non-linear} \\ k_B T & \text{linear} \end{cases}$ same as equipartition since Q_{rot} is already for "high" T

$U_{trans} = \frac{3}{2} k_B T$

} Add E_{DFT} to these terms

F = Helmholtz

$S_{gas}(T,P)$ can also be found from Q_{gas}:

$S = \underbrace{k \ln Q}_{-\frac{F}{T}} - \frac{1}{T} \frac{\partial \ln Q}{\partial \beta} = k \ln Q + \frac{U}{T}$

$\Downarrow$

$S_{vib} = \sum_i \left(\frac{-h\nu_i}{2T} - k_B \ln\left(1 - e^{-h\nu_i \beta}\right) \right) + \frac{1}{T} \sum_i \left(\frac{1}{2} h\nu_i + \frac{h\nu_i}{e^{\beta h\nu_i} - 1} \right)$

$= \int k_B \ln \left[\frac{\sqrt{\pi}}{6} \prod_{i=1}^{3} \left(\frac{2 I_i}{\hbar^2 \beta} \right)^{1/2} \right] + \frac{3}{2} k_B \rightarrow$ Non-linear

$= \int k_B \ln \left[\frac{I}{6 \beta \hbar^2} \right] + k_B \rightarrow$ linear

$S^{tot}_{trans} = Q^{tot}_{trans} = \frac{\left(Q^{moles}_{trans}\right)^{N_e}}{N!}$ (N_e = # molecules)

↖ Indistinguishable particles

$S^{tot}_{trans} = Nk \ln Q^{mol}_{trans} - k \ln N! + \frac{3}{2} k_B N$

$\approx Nk \ln Q^{mol}_{trans} - kN \ln N + kN + \frac{3}{2} k_B T$

$= Nk \ln \left[\left(\frac{m}{2\pi\beta\hbar^2} \right)^{3/2} \frac{N k_B T}{P} \frac{1}{N} \right] + \frac{5}{2} k_B T$

$\therefore \boxed{\frac{S^{tot}_{trans}}{N} = k_B \ln \left[\left(\frac{m}{2\pi\beta\hbar^2} \right)^{3/2} \frac{k_B T}{P} \right] + \frac{5}{2} k_B T \rightarrow \text{per molecule}}$

fugacities from an EOS

(iii) Real gases:

$\mu_{gas}(T,P) = \mu_{gas}(T,P^\circ) + k_B T \ln \left(\frac{f(T,P)}{f(T,P^\circ)} \right)$

↓

If P° is 1 bar, use i.g. approximations

(1) H^* on Ni(111) ; Bulk stability limit : $Ni(s) + \frac{1}{2} H_2(g) \rightleftharpoons NiH(s)$

[QE: $\frac{1}{2} H_2(g) + * \rightleftharpoons H^*$

$\frac{1}{2} \mu_{H_2(g)} = \mu_{H^*}$]

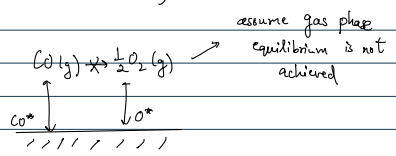
$\Delta G_{molar} = \mu_{NiH} - \frac{1}{2} \mu_{H_2} - \mu_{Ni}$

$\boxed{\Delta G_{molar} = \mu_{NiH} - \mu_H - \mu_{Ni}}$

For $\Delta G_r > 0$, $\boxed{\mu_H < \mu_{Ni:H(bulk)} - \mu_{Ni(bulk)}}$

w/ DFT $\mu_{bulk} \approx E_{DFT,\,bulk}$

(2) CO oxidation on RuO_2 (110) :



$\mu_{CO(g)} \rightleftharpoons \mu_{CO*}$

$\frac{1}{2}\mu_{O_2(g)} \approx \mu_{O*}$

Bulk stability: $Ru(s) + O_2(g) \rightleftharpoons RuO_2(s)$

$\Delta G_r = \mu_{RuO_2} - \mu_{O_2(g)} - \mu_{Ru} < 0$

$$\boxed{\frac{1}{2}\mu_{O_2(g)} = \mu_{O*} > \frac{1}{2}\left(\mu_{RuO_2(s)} - \mu_{Ru(s)}\right)}$$

Zeolites

Bulk zeolite are not equilibrium structures (bulk SiO_2 is equilibrium)
(metastable → decays at high temp)

↳ True excess quantities are not useful

$$\gamma = \frac{U_{cell} - TS - \sum \mu_i N_i - U_{clean}}{A}$$

- U_{cell}: DFT energy includes adsorbates
- TS: entropy in zeolites
- $\sum \mu_i N_i$: Not true interfacial free energy
- U_{clean}: DFT of clean zeolite
- A: Area in pores

(1) U_{cell} and U_{clean} are taken from DFT energies + Thermal corrections (as before)

(2) μ_i are evaluated similarly unless significant mass transport barries exist

(3) Entropy effects are similar (see upcoming lecture)

(4) No point in estimating bulk stability ranges

(5) If pores have wetting/non-wetting transition, both energies and entropies are best evaluated using Ab-initio molecular dynamics

Dealing w/ nanoparticles → need to know what the surface looks like:

1) Simple case: surface energy in a vacuum

$$\gamma = \left(U^{KS} - TS^{KS} - \sum_i \mu_i N_i^{KS} \right) \frac{1}{A}$$

- $U^{KS} = U_{\underset{DFT}{surf}} - \sum_{metal} \mu_{\underset{bulk}{metal}} N_{\underset{DFT}{metal}}$

For an ordered surface; $N^{KS}_{metal} \approx 0$

$$\therefore \quad \gamma = \frac{1}{A}\left(U_{\underset{DFT}{surf}} - \cancel{TS^{KS}}^{\,\to 0} - \sum \mu_{metal} N_{metal} \right)$$

(N_{metal}: total)

Typically, $\mu_{metal} = g_{metal} \approx e^{\underset{DFT}{bulk}}_{metal}$

(= DFT energy per bulk metal atom)